

Can metadata identify highly rated apps?

Existing US App Store apps

Class 1: rating ≥ 4.5

Class 0: rating < 4.5

**2,029
eligible
apps**

Seven metadata inputs · Four classifiers · Fixed holdout

Data source and collection conditions

Apple iTunes Search API · US storefront

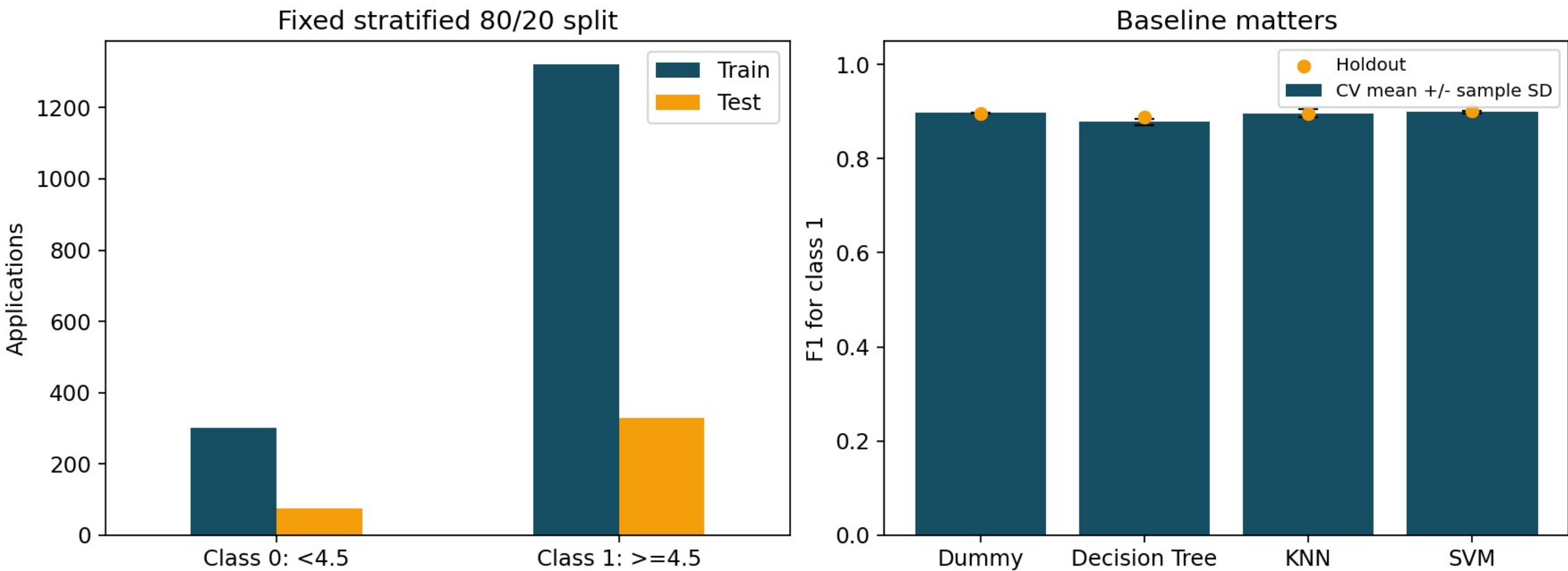
17 requests used; 132 keywords planned

Pause ≥ 3.2 seconds; cache raw responses privately

Latest observation per trackId; at least 50 ratings

robots.txt disallows /search* • Raw JSON/descriptions stay private

Cleaning and class balance



Raw rows: 2,763 → eligible unique apps: 2,029

Seven inputs; rating fields are excluded

price_usd · size_mb · primary_genre

content_advisory_rating · language_count

release_year · update_recency_days

averageUserRating → target only

userRatingCount → ≥ 50 eligibility filter only

Validation without data leakage

Stratified 80/20 split · random_state = 42

Five-fold stratified CV within train

Fold-local imputation → encoding/scaling → model

Dummy · Decision Tree · KNN · SVM

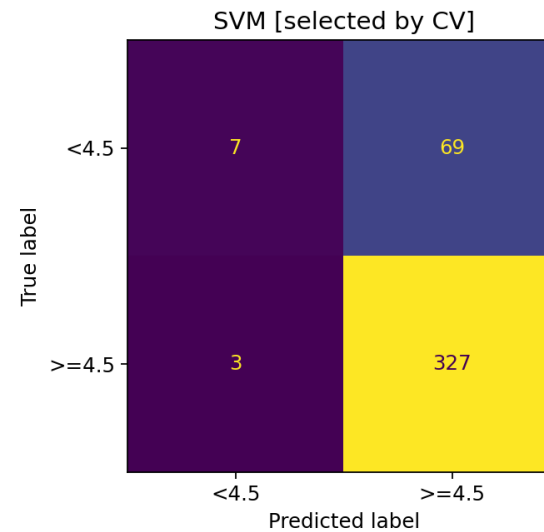
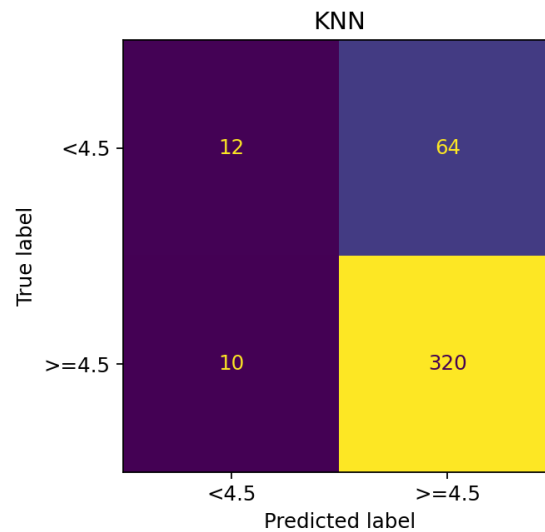
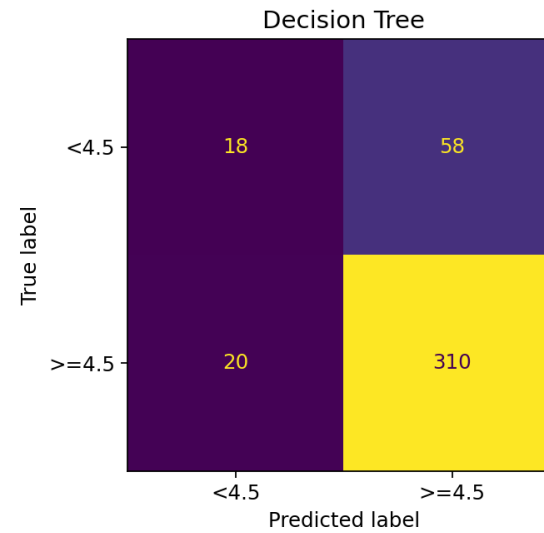
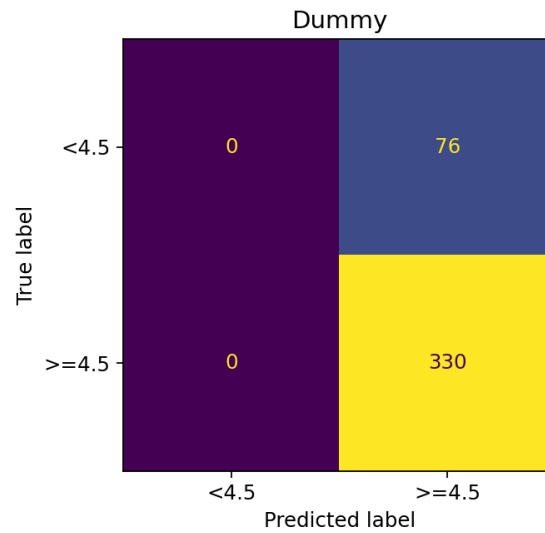
Selected by mean CV F1: SVM

Results: F1 needs a baseline

Model	CV F1 $\pm$ SD	Test F1	AUC	Precision	Recall
Dummy	0.897 $\pm$ 0.001	0.897	0.500	0.813	1.000
Decision Tree	0.878 $\pm$ 0.007	0.888	0.678	0.842	0.939
KNN	0.896 $\pm$ 0.009	0.896	0.639	0.833	0.970
SVM	0.899 $\pm$ 0.003	0.901	0.620	0.826	0.991

SVM: holdout F1 0.901 vs Dummy 0.897 (gain +0.004); ROC-AUC 0.620. It detected 7/76 lower-rated apps. The F1 gain is descriptive; no statistical significance is claimed.

Where predictions fail



Real cases, anonymized

Case 1 · FP · true 0, predicted 1

True rating is within 0.1 of the 4.5 boundary

Case 4 · FN · true 1, predicted 0

Metadata alone does not explain the observed rating

Full private audit is generated locally; public cases omit IDs, names and exact ratings.

Conclusion and next steps

SVM: holdout F1 0.901 vs Dummy 0.897 (gain +0.004); ROC-AUC 0.620. It detected 7/76 lower-rated apps. The F1 gain is descriptive; no statistical significance is claimed.

Limits: search-ranking bias, one snapshot, developer overlap

No evidence for pre-launch success prediction

Final: neural network · text analysis · Streamlit